# EXPERIMENT 5 – INFORMATION EXTRACTION USING spaCy

## AIM

To extract tokens, lemmas, POS tags and named entities from customer reviews using spaCy.

## ALGORITHM

1. Load and clean up to 10,000 review texts.
2. Load the English spaCy model.
3. Display tokens, lemmas and POS tags for five reviews.
4. Count POS categories and collect named entities with labels.
5. Create and test an extraction function on new reviews.

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import pandas as pd, spacy
from collections import Counter
import csv
class="cm"># If needed: python -m spacy download en_core_web_sm
nlp=spacy.load(class="s">"en_core_web_sm")
class="cm"># Added quoting=csv.QUOTE_NONE and on_bad_lines=class="s">'skip' to
df=pd.read_csv(class="s">"Reviews.csv", quoting=csv.QUOTE_NONE, on_bad_lines=cl
pos=Counter(); entities=[]
for text in df.Text.head(5):
 d=nlp(str(text)[:5000]); print(class="s">"REVIEW:",text); print(class="s">"Tok
for text in df.Text:
 d=nlp(str(text)[:5000]); pos.update(t.pos_ for t in d)
 for e in d.ents: entities.append({class="s">"Review":str(text)[:100],class="s"
print(class="s">"POS frequencies:",pos); print(pd.DataFrame(entities).head(20))
def extract_info(text):
```

```
REVIEW: I have bought several of the Vitality canned dog food products and h
Tokens: ['I', 'have', 'bought', 'several', 'of', 'the', 'Vitality', 'canned'
Lemmas: ['I', 'have', 'buy', 'several', 'of', 'the', 'Vitality', 'can', 'dog
POS: [('I', 'PRON'), ('have', 'AUX'), ('bought', 'VERB'), ('several', 'ADJ')
REVIEW: "Product arrived labeled as Jumbo Salted Peanuts...the peanuts were
Tokens: ['"', 'Product', 'arrived', 'labeled', 'as', 'Jumbo', 'Salted', 'Pea
Lemmas: ['"', 'product', 'arrived', 'label', 'as', 'Jumbo', 'Salted', 'Peanu
POS: [('"', 'PUNCT'), ('Product', 'NOUN'), ('arrived', 'AUX'), ('labeled', '
REVIEW: If you are looking for the secret ingredient in Robitussin I believe
Tokens: ['If', 'you', 'are', 'looking', 'for', 'the', 'secret', 'ingredient'
Lemmas: ['if', 'you', 'be', 'look', 'for', 'the', 'secret', 'ingredient', 'i
POS: [('If', 'SCONJ'), ('you', 'PRON'), ('are', 'AUX'), ('looking', 'VERB'),
REVIEW: Right now I'm mostly just sprouting this so my cats can eat the gras
Tokens: ['Right', 'now', 'I', "'m", 'mostly', 'just', 'sprouting', 'this', '
Lemmas: ['right', 'now', 'I', 'be', 'mostly', 'just', 'sprout', 'this', 'so'
POS: [('Right', 'ADV'), ('now', 'ADV'), ('I', 'PRON'), ("'m", 'AUX'), ('most
REVIEW: This is a very healthy dog food. Good for their digestion. Also good
Tokens: ['This', 'is', 'a', 'very', 'healthy', 'dog', 'food', '.', 'Good', '
Lemmas: ['this', 'be', 'a', 'very', 'healthy', 'dog', 'food', '.', 'good', '
POS: [('This', 'PRON'), ('is', 'AUX'), ('a', 'DET'), ('very', 'ADV'), ('heal
POS frequencies: Counter({'NOUN': 76347, 'PRON': 59809, 'VERB': 54250, 'PUNC
                                                Review  \
0   I have bought several of the Vitality canned d...
1   I have bought several of the Vitality canned d...
2   "Product arrived labeled as Jumbo Salted Peanu...
3   "Product arrived labeled as Jumbo Salted Peanu...
4   If you are looking for the secret ingredient i...
5   If you are looking for the secret ingredient i...
6   Right now I'm mostly just sprouting this so my...
7   good flavor! these came securely packed... the...
8   The Strawberry Twizzlers are my guilty pleasur...
9   The Strawberry Twizzlers are my guilty pleasur...
10  The Strawberry Twizzlers are my guilty pleasur...
11  I am very satisfied with my Twizzler purchase....
12  I can remember buying this candy as a kid and ...
13  I was so glad Amazon carried these batteries. ...
14  I have McCann's Oatmeal every morning and by o...
15  I have McCann's Oatmeal every morning and by o...
16  I have McCann's Oatmeal every morning and by o...
```

```
                      Entity        Label
0                   Vitality          ORG
1                 My Labrador       PERSON
2                Jumbo Salted          ORG
3                       Jumbo  WORK_OF_ART
4                  Robitussin          GPE
5    the Root Beer Extract I           LAW
6                  Wheatgrass       PERSON
7                   Twizzlers          ORG
8   The Strawberry Twizzlers          ORG
9                  Six pounds     QUANTITY
10                         I.          ORG
11                  Twizzler          ORG
12           all these years         DATE
13                    Amazon          ORG
14                    McCann          ORG
15              every morning         TIME
16                    Amazon          ORG
17               almost $3.00        MONEY
18                    McCann          ORG
19                south texas          LOC
{'Tokens': ['The', 'product', 'arrived', 'in', 'Chennai', 'on', 'Monday', '.
{'Tokens': ['I', 'paid', '25', 'dollars', 'for', 'this', 'coffee', '.'], 'Le
{'Tokens': ['Acme', 'Foods', 'released', 'a', 'product', 'in', '2024', '.'],
```

## RESULT

Thus, spaCy extracts linguistic information and named entities from unstructured customer reviews.